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Homework 1
CS 336

- Problem 1 Consider the following bank database with 6 tables. For each table, the primary key is underlined.

$Table$	$Attributes$
$branch$	$(\underline{branch\_name}, branch\_city, assets)$
$customer$	$(\underline{ID}, customer\_name, street, city)$
$loan$	$(\underline{loan\_number}, branch\_name, amount)$
$borrower$	$(\underline{CustomerID, loan\_number})$
$account$	$(\underline{account\_number}, branch\_name, balance)$
$depositor$	$(\underline{CustomerID, account\_number})$

1. *Find the ID and name of customers who live in "Boston".*
   $\Pi_{ID,customer\_name}(\sigma_{city=\text{"}Boston\text{"}}(customer))$
2. *Find the ID of each borrower who has a loan in branch "Edison."*
   $\Pi_{CustomerID}(\sigma_{branch\_name=\text{"}Edison\text{"}}(borrower \bowtie_{borrower.loan\_number=loan.loan\_number} loan))$
3. *Find the ID and names of customers who have deposits in "Edison" branch but do not have loans in "Edison" branch.*
   $\Pi_{CustomerID}(\sigma_{branch\_name=\text{"}Edison\text{"}})-$
   $\Pi_{ID,customer\_name}(\sigma_{branch\_name=\text{"}Edison\text{"}}($
   $customer \bowtie_{customer.ID=borrower.CustomerID} borrower \bowtie_{borrower.loan\_number=loan.loan\_number}$
   $loan))$

- Problem 2

1.